FRAME WORK FOR THE SMART HOME AUTOMATION IN INTEGRATION WITH CLOUD

CAPSTONE PROJECT

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Computer Applications

in

Department of Computer Applications

by

MAHESH KUMAR.M-22BCA0238

HARI PRASATH.M-22BCA0241

Under the guidance of

Dr.Jothish Kumar M

School of Computer Science Engineering and Information Systems

VIT, Vellore

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

April, 2025

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DECLARATION

I hereby declare that the - Capstone Project entitled **"Frame Work For The Smart Home Automation In Integration With Cloud"** submitted by me, for the award of the degree of Bachelor of Computer Applications in Department of Computer Applications, School of Computer Science Engineering and Information Systems to VIT is a record of bonafide work carried out by me under the supervision of Dr .JOTHISH KUMAR.M.Assistant Professor Sr. Grade 2, SCORE, VIT, Vellore.

I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place: Vellore

Date: 05-04-2025

maha

M. Haniferth

Signature of the Candidate

CERTIFICATE

This is to certify that the **Capstone Project** entitled **"Frame Work For The Smart Home Automation In Integration With Cloud"** submitted by **MAHESH KUMAR M(22BCA0238), HARI PRASATH M(22BCA0241)**, SCORE, VIT, for the award of the degree of Bachelor of Computer Applications in Department of Computer Applications, is a record of bonafide work carried out by him / her under my supervision during the period, 13. 12. 2024 to 17.04.2025, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute orany other institute or university. The Capstone project fulfills the requirements and regulations ofthe University and in my opinion meets the necessary standards for submission.

Place: Vellore
Date: 05-04-2025

Signature of the VIT-SCORE - Guide

Internal Examiner

External Examiner

Head of the Department
Department of Computer Applications

ACKNOWLEDGEMENT

It is my pleasure to express with a deep sense of gratitude to my Capstone project guide Dr. JOTHISH KUMAR M, Assistant Professor Sr. Grade 2, School of Computer Science Engineering and Information Systems, Vellore Institute of Technology, Vellore for his/her constant guidance, continual encouragement, in my endeavor. My association with him/her is not confined to academics only, but it is a great opportunity on my part to work with an intellectual and an expert in the field of Internet of Things.

"I would like to express my heartfelt gratitude to Honorable Chancellor **Dr. G Viswanathan**; respected Vice Presidents **Mr. Sankar Viswanathan**, **Dr. Sekar Viswanathan**, Vice Chancellor **Dr. V. S. Kanchana Bhaaskaran**; Pro-Vice Chancellor **Dr. Partha Sharathi Mallick**; and Registrar **Dr. Jayabarathi T**.

My whole-hearted thanks to Dean **Dr. Daphne Lopez**, School of Computer Science Engineering and Information Systems, Head, Department of Computer Applications **Dr. E Vijayan**, BCA Project Coordinator **Dr. Senthil Kumar T**, SCORE School Project Coordinator **Dr. Thandeeswaran R**, all faculty, staff and members working as limbs of our university for their continuous guidance throughout my course of study in unlimited ways.

It is indeed a pleasure to thank my parents and friends who persuaded and encouraged me to take up and complete my capstone project successfully. Last, but not least, I express my gratitude and appreciation to all those who have helped me directly or indirectly towards the successful completion of the capstone project.

Place: Vellore

Date:

Students Name: MAHESH KUMAR M(22BCA0238),
HARI PRASATH M(22BCA0241)

Executive Summary

The objective of the project is to develop a cloud-based smart home automation system with remote control and monitoring through a mobile application by utilizing a mobile application using ESP8266 microcontroller form device interfaces and real-time connectivity with Firebase and Blynk Cloud.

The system provides advanced accessibility, security, and automation. With movement from Bluetooth-based (HC-05) to cloud and Wi-Fi, the solution now has plug-and-play anywhere with real-time location tracking, scheduling of automation, and AI-based decision-making. Smartphone app supports basic interface, voice, push notification, and sensor-input wise smart automation. Security is provided through authentication mechanisms, end-to-end encrypted chat, and cloud logging through secure and stable user experience. Moreover, the system is scalable and extensible for new automation features and new devices.

The project will develop a smart, hi-tech, power-efficient home system with energy usage optimized, maximum security, and providing users responsive and easy home experience.

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CHAPTER 1

INTRODUCTION

The fast growth of Internet of Things (IoT) and cloud computing has transformed home automation into an intelligent, more networked, and automated experience, replacing the traditional household management. Smart home automation systems allow customers to monitor, control, and automate devices at home remotely using cloud-based interfaces and mobile apps. This integration not only improves convenience and security but also energy efficiency and cost savings.

Wired systems used to form the basis of traditional home automation. They were cumbersome and costly. With the advent of wireless IoT-based systems, smart home automation has become economical, scalable, and universal.ESP8266 microcontrollers enable real-time interaction between cloud servers and household appliances, facilitating easy automation. Cloud platforms such as Blynk Cloud and Firebase provide data storage, synchronization, and remote access, which enable users to remotely monitor and control their smart home.

This project has envisioned a smart home automation design with cloud computing integration to improve the efficiency, security, and ease of use of home automation systems. With IoT sensors, actuators, and cloud services, the design will offer real-time monitoring, automated decision-making, and remote control. A mobile app will also be created so that users will have an easy-to-use interface to manage their smart home appliances.

1.1 Objective

- ✓ To create an intelligent home-based home automation system capable of doing cloud computing for real-time monitoring, control, and management of home appliances.
- ✓ To integrate smart devices and sensors (e.g., fans, lights, temperature sensors, relays) into a central microcontroller (e.g., ESP8266 or Arduino) to control devices automatically.
- ✓ To facilitate remote control and access of home appliances via web or mobile applications on cloud-based communication platforms (e.g., Firebase, AWS, or MQTT).
- ✓ To facilitate secure data sharing and device access via authentication mechanisms and encryptionbetween cloudand device.
- ✓ To facilitate increased energy efficiency and convenience through injecting automation rules (e.g., schedule-control, sensor-evoked events) into the cloud-based platform.
- ✓ For statistics to provide information on usage and processing of smart devices via the cloud in an effort to make recommendations and provide insights on efficient energy usage.

1.2 Motivation

There is increased need for intelligent, energy-efficient, and secure home automation systems. Some of the most important drivers include:

- ✓ Remote Access and Control

  Manually controlled and locally accessed conventional home automation systems. With the integration of cloud computing into the equation, customers are able to have remote access and control of home appliances through mobile applications.

- ✓ Energy Efficiency

  Increased energy prices and weather conditions have required the use of energy-saving smart systems. Real-time monitoring and automation are incorporated in the project to prevent wastage of energy usage.

✓ Safety and Security Enhanced

Home security safety concerns like intruder, fire, and gas leak require real-time monitoring systems. The system implemented in the project incorporates intelligent monitoring, motion detection, and alarm signaling to enhance security of the home.

✓ Scalability and Customization

Today's home automation systems are not typically interoperable and scalable. It is simple to scale out a cloud system, and several sensors and devices can be installed, and customized automation rules can be provided based on the user's need

1.3 Background

Home automation has emerged following the wide applicability of Internet of Things (IoT) and cloud computing technologies globally. Communication earlier occurred using wired-based technology as a previous solution in home automation that was costly to install and cumbersome to achieve. However, using wireless IoT-based technology, intelligent homes can currently accommodate remote control, wise automation, as well as energy-saving control of appliances and equipment.

Smart home automation systems cloud-based enable users to control and monitor domestic appliances remotely through cellular applications or internet interfaces. Smart home automation systems cloud-based enable wireless microcontrollers like ESP8266, which communicate with heaps of sensors, actuators, and cloud programs to deliver instantaneous automation. Tools like Firebase and Blynk Cloud offer safe storage capacity, device synchronization, and real-time access to information on the move, thus making home automation systems efficient and reliable.

As there is a increasing need for automation, security, and energy efficiency, application of cloud computing integration with home automation is a feasible, economical, and intelligent solution. The project here discussed suggests a cloud-based automated smart home system allowing for easy remote monitoring and automation of domestic appliances with maximum security and maximum energy efficiency.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

The "Framework for Smart Home Automation in Integration with Cloud" is a project aimed at developing a comprehensive and scalable system that automates and remotely manages home appliances using Internet of Things (IoT) technologies and cloud computing. The framework focuses on enhancing user convenience, security, and energy efficiency by allowing centralized control of household devices via mobile or web interfaces.

In this project, various electronic components such as microcontrollers (e.g., Arduino Uno or ESP8266), sensors, relays, and modules like HC-05 Bluetooth and Wi-Fi are used to build a smart environment within a home. These devices communicate with each other and with the cloud to provide seamless, real-time monitoring and control. The cloud platform stores and processes the data collected from sensors and allows users to access this data and control appliances remotely.

Key features of the framework include device status monitoring, scheduling operations, sensor-based automation (e.g., turning on lights when motion is detected), and user notifications. Cloud integration ensures that data is not only stored securely but also accessible globally, making the smart home system responsive and reliable from any location.

Furthermore, the system is designed to be scalable, allowing the integration of additional devices or services in the future without significant changes to the core architecture. Emphasis is also placed on data security and user authentication to protect personal and device information.

By implementing this framework, the project demonstrates how modern IoT and cloud technologies can be harmoniously integrated to build an efficient, smart, and user-friendly home automation system.

2.1 Goals

Smart Home Automation System will offer users a cost-effective and editable method of remote control of home appliances through Wi-Fi (ESP8266), Bluetooth (HC-05), and cloud networking (Firebase & Blynk Cloud). The system will lead to increased convenience, efficiency, and security through IoT-based automation. Of the primary goals is the offering of remote access through Firebase Cloud and Blynk Cloud, offering remote monitoring and control of devices. Blynk Cloud also supports delivery of an IoT device management system that is easy to control, e.g., real-time alerting, graphical user interface dashboard, and easy interfacing with ESP8266. Control by the local Bluetooth with the HC-05 module also supports Wi-Fi-less operation.

The system is cost-effective with remote on/off of devices while in stand-by mode, hence saving electricity and cost. Manual switches are incorporated into a stand-by control system, which will act as a back-up in case of system failure. The project has been modularly designed such that facility for future upgradation by using other devices and sensors is available. Security is also maximum with password-based mobile application software and encryption of data communication which makes the system secure against misuse.

Apart from that, the system provides an uninterrupted Wi-Fi to Bluetooth mode conversion without disturbing the continuity of usage even during connection. Cost-saving is also provided topmost priority by utilizing ultra-low-cost microcontrollers, relays, and sensors with high performance. With the help of Firebase, Blynk Cloud, ESP8266, and HC-05, the project attempts to provide smart, scalable, and user-friendly home automation system to make life easier day by day while enhancing efficiency in both directions of energy as well as security.

CHAPTER 3

TECHNICAL SPCIFICATION

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3.1 Hardware Specifications

Basic Components

- ✓ **microcontroller**: Arduino Uno (ATmega328P)
- ✓ **Wi-Fi Module**: ESP8266 (for cloud-based remote control)
- ✓ **Bluetooth Module**: HC-05 (for local control)
- ✓ **Relay Module**: 2-Channel/4-Channel Relay (to turn appliances ON/OFF)
- ✓ **Power Supply**: 9V – 12V Battery / 5V USB Adapter
- ✓ **Switches**: Manual switches for direct appliance control
- ✓ **Jumper Wires**: For connecting components
- ✓ **PCB or Breadboard**: For assembling circuit
- ✓ **Light Bulb / LED Lamp**: To test automation
- ✓ **Electric Fan**: Controlled via relay for home automation

3.2 Software Requirements

Basic Software

- Arduino IDE – For programming and uploading the program to Arduino
- C/C++ (Arduino Language) – For programming the microcontroller
- Blynk IoT App – For cloud-based remote monitoring and control
- Basic Communication Libraries
- ESP8266WiFi Library – Supports Wi-Fi communication
- Blynk Library – For communication with Blynk Cloud
- SoftwareSerial Library – For Bluetooth-based communication

CHAPTER 4

DESIGN APPROACH AND DETAILS

Smart Home Automation System follows a modular design approach based on Wi-Fi (ESP8266), Bluetooth (HC-05), and Blynk Cloud to control home appliances. It is executed in three control modes:

- ✓ **Cloud-Based Remote Control** – ESP8266 and Blynk Cloud-based remote control using a smartphone app.
- ✓ **Bluetooth-Based Local Control** – HC-05-based local appliance control without depending on the internet.
- ✓ **Manual Control** – Availability of manual switches for direct appliance control.

The system utilizes a relay module to carry out electric device ON/OFF switching through the reception of commands via a mobile application. The mobile application offers support for a Blynk Cloud communication for real-time update and control. The input signals are handled by the Arduino Uno microcontroller, while the relays are controlled accordingly.

Materials & Methods

- Hardware Components:
- Microcontroller: Arduino Uno
- Wireless Modules: ESP8266 (Wi-Fi), HC-05 (Bluetooth)
- Actuators: Relay Module (2/4 channels)
- Power Source: 9V-12V battery / 5V adapter
- Output Devices: Light, Fan

- Manual Controls: Physical switches for direct control
- Arduino IDE for programming and uploading codes
- Blynk Cloud for IoT-based monitoring and control
- MIT App Inventor / Android Studio for mobile app development
- ESP8266WiFi and Blynk libraries for connectivity

The combination of the above factors provides an efficient, economical, and scalable home automation system.

4.2 Codes and Standards

- The following technical standards for reliability, safety, and compliance are utilized in the project as mentioned below:
- IEEE 802.11 (Wi-Fi Standard) – Offers secure wireless communication with ESP8266.
- IEEE 802.15.1 (Bluetooth Standard) – Offers a low-power short-range communication protocol for HC-05.
- IEC 60669 (Switching Devices Standard) – Offers electrical switching safely using relays.
- IEC 60364 (Electrical Safety Standards) – Offers safe circuit design and eliminates hazards.
- Arduino Coding Standards – Complies with proper indentation, naming, and structured programming for maintenance ease.

4.3 Constraints, Alternatives, and Tradeoffs

✓ **Constraints:**

Constraints, Alternatives, and Tradeoffs are key system design factors.

System constraints are Internet Dependency, where there must be an ongoing Wi-Fi connection to operate remotely with Blynk Cloud.

Power Supply Limitation, where relays require a continuous power supply and variation in voltage will affect performance. Range Limitation is also a constraint, where HC-05 Bluetooth has a range of 10 meters and therefore local control is limited.

Besides that, Security Risks also occur due to unauthenticated access risk since authentication does not exist in cloud control.

- ✓ **Alternative & Tradeoffs:**

- ✓ There are numerous substitutes for these constraints. For instance, substituting for Wi-Fi (ESP8266) is mobile network-based control using a GSM Module (SIM800L).
- ✓ Another substitute is RF Modules (NRF24L01) in place of Bluetooth Alternate (HC-05) for increased range.

- ✓ Blynk Cloud can be replaced with MQTT protocol by a private cloud server for enhanced security.

- ✓ Also, Arduino Uno can be replaced with ESP32, which supports onboard Wi-Fi and Bluetooth, reducing the use of external modules.

- ✓ Tradeoffs cannot be avoided while deciding between two items. One such example is Wi-Fi or Bluetooth where Wi-Fi provides remote control but with internet availability, and Bluetooth provides offline control but limited range.

- ✓ A sample is ESP8266 or ESP32, with ESP8266 being low-cost but does not have internal Bluetooth and ESP32 does not have internal Bluetooth and Wi-Fi but is high-cost.
- ✓ Another trade-off is Blynk Cloud vs. MQTT, which is pre-configured but internet-based, while MQTT is flexible but needs to be configured.

- ✓ Last but not least, Manual Switches vs. Full Automation is a trade-off, while manual switches are reliable but inefficient compared to full automation.

4.4 System Architecture

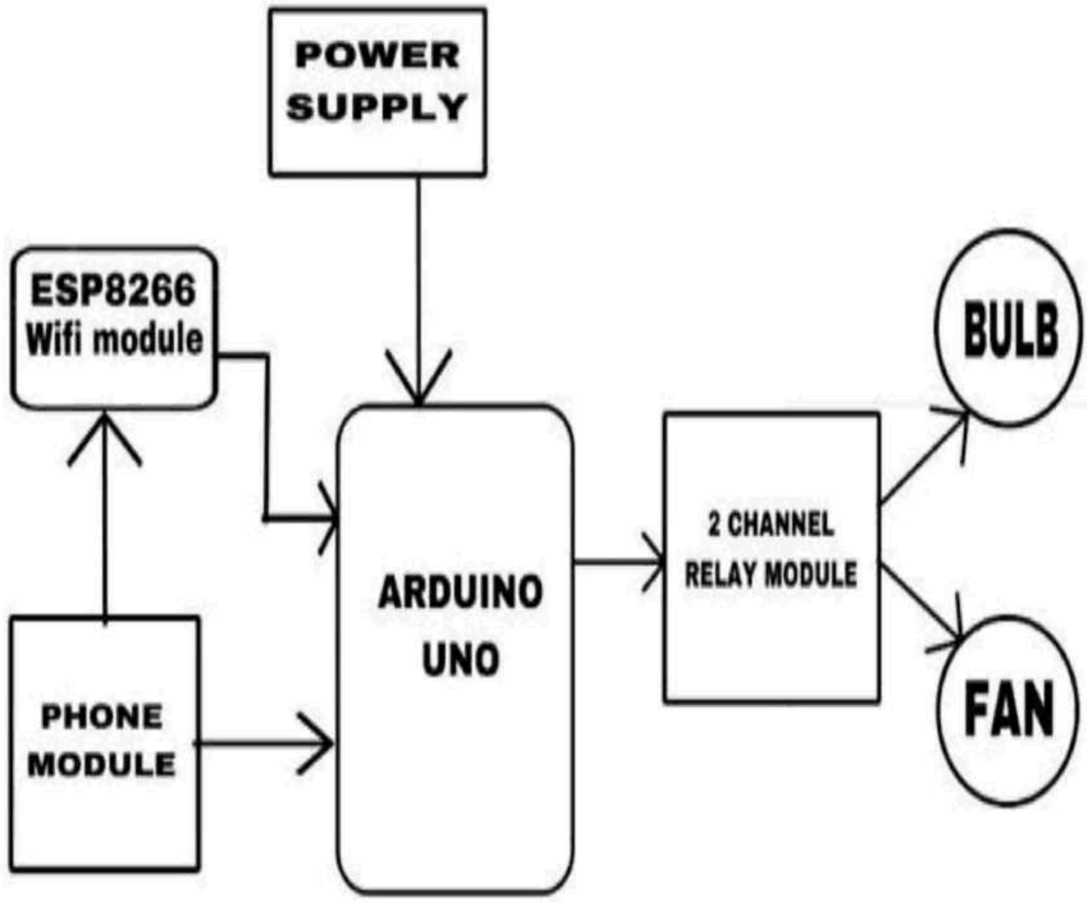

Fig.4.4.1 System Architecture

4.5 Class Diagram

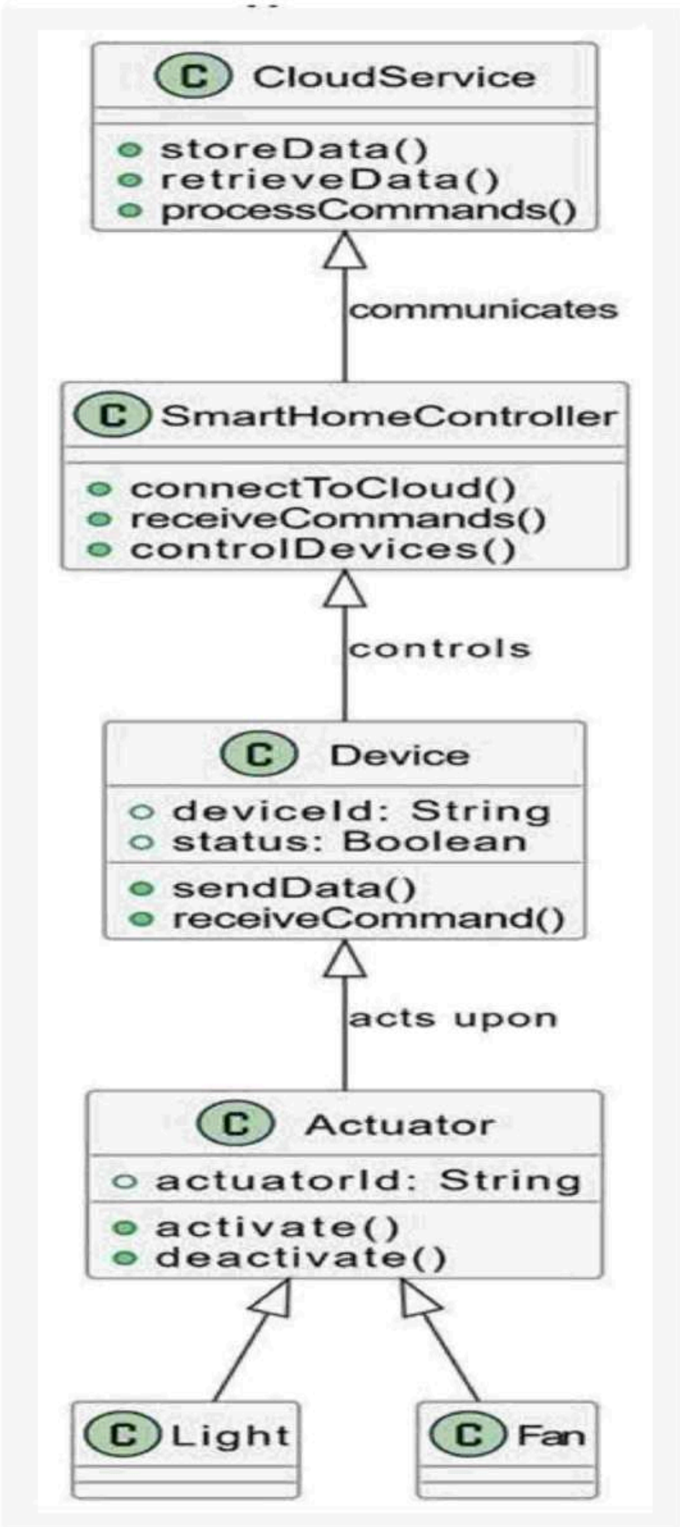

Fig.4.5.1 Class Diagram

4.7 Use case Diagram

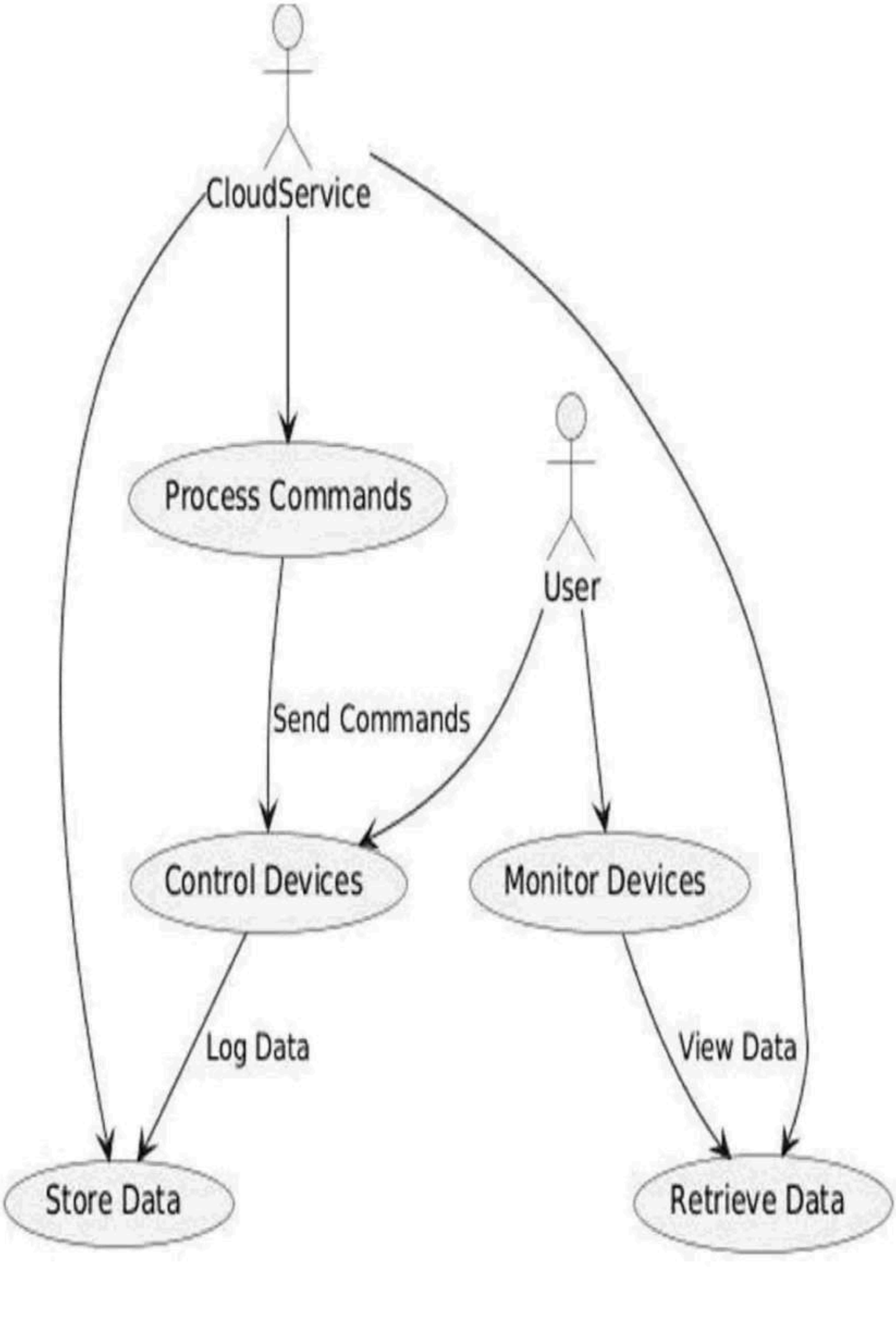

Fig.4.7.1 Use Case Diagram

CHAPTER 5

SCHEDULE, TASKS AND MILESTONES

Table 5.1: Project Phases, Tasks.

Phase	Tasks	Duration
Phase 1: Planning & Research	In the initial phase, we studied elementary technologies used in home automation such as Wi-Fi, Bluetooth, and cloud services such as Blynk. We studied different control mechanisms and how to utilize each one of them. Throughout the learning process, all materials of hardware required such as Arduino boards, ESP8266 modules, HC-05 Bluetooth modules, relays, switches, and power supplies were purchased to design a platform for upcoming phases.	Week 1 – Week 2
Phase 2:Hardware Installation & Testing	Here, we finished the assembly of the actual circuitry. We mounted the circuitry correctly on a breadboard and initiated the soldering process wherever required. All and every one of	Week 3 – Week 4

	the components such as switches, relays, and both wireless modules (Wi-Fi & Bluetooth) were individually tested to see if they were working or not before assembling.	
Phase 3: Software Development	Parallel to the installation of hardware, we moved to software installation. For remote as well as local control of the appliances using Wi-Fi and Bluetooth, Arduino code was utilized. Concurrently, we started installation of local and remote control via the services of Blynk Cloud. Installation of a classy user interface for controlling the smart home system was our most significant task.	Week 5–Week 6
Phase 4: System Integration & Testing	We implemented hardware and software as one system here. We minimized response time of Wi-Fi and Bluetooth commands, codified bugs, and optimized performance in general. There was extensive testing to ensure stability, reliability, and real-time	Week 7–Week 8

	control on all devices involved.	
Phase 5: Final Deployment & Demonstration	Finally, at the final stage, the system was placed in a compact sealed enclosure to demonstrate and shield it. We performed full-scale testing and demonstration of the project, its running, modes of control, and response in real-time.	Week 9– Week 10

Table 5.2: Milestones.

Milestones	Expected Completion
Research and purchase of components complete	Week 2
Component testing and hardware installation complete	Week 4
Software and mobile application development are complete	Week 6
System integration and functional testing are complete	Week 8
Final project is deployed and demonstrated successfully	Week 10

CHAPTER 6

PROJECT DEMONSTRATION

6.1 Sample Codes

Code For Cloud

```
void(* resetFunc) (void) = 0;
//resetFunc();
#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>
#include <ESP8266WebServer.h>
#include <EEPROM.h>
int count =0;
//Variables
int i = 0;
int statusCode;
const char* ssid = "Default_SSID";
const char* passphrase = "Default_Password";
String st;
String content;
#define LED  0
//Function Decalration
bool testWifi(void);
void launchWeb(void);
void setupAP(void);

//Establishing Local server at port 80 whenever required
ESP8266WebServer server(80);
#define BLYNK_TEMPLATE_ID "TMPL34Gng7PQb"
```

```
#define BLYNK_TEMPLATE_NAME "smart switch"
#define BLYNK_AUTH_TOKEN "HiSflOgGiS1sccl4quGlWCSvRjKSMY0j"
/////////////////////////////////////////////////////////////////////////////
/*#define BLYNK_TEMPLATE_ID "TMPL3cAk-6C3I"
#define BLYNK_TEMPLATE_NAME "smart wifi"
#define BLYNK_AUTH_TOKEN "jf8QN4F8I4Lsz8SbjsxZWk95xyNDrIpJ"
*/
bool fetch_blynk_state = false;

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#define RelayPin1 4  //GPIO-0
#define VPIN_BUTTON_1    V1
bool toggleState_1 = LOW; //remember the toggle state for relay
#define RelayPin2 16  //GPIO-2
#define VPIN_BUTTON_2    V2
bool toggleState_2 = LOW;
char auth[] = BLYNK_AUTH_TOKEN;
BlynkTimer timer;
/////////////////////////////////////////////////////////////////////////////
BLYNK_WRITE(VPIN_BUTTON_1)
{
 toggleState_1 = param.asInt();
 digitalWrite(RelayPin1, !toggleState_1);
 delay(1000);
}
BLYNK_WRITE(VPIN_BUTTON_2)
{
 toggleState_2 = param.asInt();
 digitalWrite(RelayPin2, !toggleState_2);
 delay(1000);
}
```

```
BLYNK_CONNECTED() {
 // Request the latest state from the server
 if (fetch_blynk_state){
 Blynk.syncVirtual(VPIN_BUTTON_1);
 Blynk.syncVirtual(VPIN_BUTTON_2);
 }
 else{
  Blynk.virtualWrite(VPIN_BUTTON_1, toggleState_1);
  Blynk.virtualWrite(VPIN_BUTTON_2, toggleState_2);
 }
}



////////////////////////////////////////////////////////////////////////////////////////////////
void setup()
{

 Serial.begin(115200); //Initialising if(DEBUG)Serial Monitor
 Serial.println();
 Serial.println("Disconnecting current wifi connection");
 WiFi.disconnect();
 EEPROM.begin(512); //Initialasing EEPROM
 delay(10);
 pinMode(LED, OUTPUT);
 digitalWrite(LED, HIGH);
 Serial.println();
```

```
Serial.println();
Serial.println("Startup");

//-------------------------- Read eeprom for ssid and pass
Serial.println("Reading EEPROM ssid");

String esid;
for (int i = 0; i < 32; ++i)
{
 esid += char(EEPROM.read(i));
}
Serial.println();
Serial.print("SSID: ");
Serial.println(esid);
Serial.println("Reading EEPROM pass");

String epass = "";
for (int i = 32; i < 96; ++i)
{
 epass += char(EEPROM.read(i));
}
Serial.print("PASS: ");
Serial.println(epass);


WiFi.begin(esid.c_str(), epass.c_str());
if (testWifi())
{
 Serial.println("Succesfully Connected!!!");
 /////////////////////////////////////
 pinMode(RelayPin1, OUTPUT);
 pinMode(RelayPin2, OUTPUT);
```

```
 digitalWrite(LED,HIGH);
//During Starting all Relays should TURN OFF
 digitalWrite(RelayPin1, !toggleState_1);
 digitalWrite(RelayPin2, !toggleState_2);
//WiFi.begin(ssid, pass);
Blynk.config(auth);
delay(1000);
if (!fetch_blynk_state){
 Blynk.virtualWrite(VPIN_BUTTON_1, toggleState_1);
 Blynk.virtualWrite(VPIN_BUTTON_2, toggleState_2);
}
 ///////////////////////////////////////////////////////
 return;
}
else
{
 Serial.println("Turning the HotSpot On");
 launchWeb();
 setupAP();// Setup HotSpot
 digitalWrite(LED,LOW);
 WidgetLED led1(V3);
 led1.off();
}

Serial.println();
Serial.println("Waiting.");

while ((WiFi.status() != WL_CONNECTED))
{ count ++;
 Serial.print(".");
 delay(100);
 server.handleClient();
```

```
   if(count>1800){resetFunc();}
  }

}
void loop() {
 if ((WiFi.status() == WL_CONNECTED))
 {
  WidgetLED led1(V3);
  led1.on();
   bool l = digitalRead(D6);
   Blynk.run();
   timer.run();
 if(l==LOW){WidgetLED led2(V4); led2.on();}else{WidgetLED led2(V4); led2.off();}
 if (!Blynk.connected()) {
   toggleState_1 = LOW;
   digitalWrite(RelayPin1, !toggleState_1);
   toggleState_2 = LOW;
   digitalWrite(RelayPin2, !toggleState_2);
  }


  }
}
```

Code For Bluetooth:

```
// Arduino 2-Channel Bluetooth Automation Code
//jmdelectroniks//90800265658/vellore
#include <SoftwareSerial.h>

// Bluetooth module connections
const int RXPin = 2; // Connect Bluetooth TX to Arduino RX pin 2
const int TXPin = 3; // Connect Bluetooth RX to Arduino TX pin 2

SoftwareSerial bluetooth(RXPin, TXPin); // RX, TX
int esp_light =A2;
int esp_fan = A3;
int mode=0;
// Relay or LED pins
const int fan_pin = A5;
const int light_pin = 11;

void setup() {
  pinMode(fan_pin, OUTPUT);
  pinMode(light_pin, OUTPUT);
  pinMode(esp_light,INPUT);
  pinMode(esp_fan, INPUT);
  digitalWrite(fan_pin, LOW); // Initialize relays to OFF
  digitalWrite(light_pin, LOW);
  pinMode(A4, INPUT_PULLUP);
  Serial.begin(9600);
  bluetooth.begin(9600);
  pinMode(13, OUTPUT);
  Serial.println("Bluetooth 2-Channel Automation Ready!");
  bluetooth.println("Bluetooth 2-Channel Automation Ready!"); //send to bluetooth too.
```

```
}

void loop()
{
int state=digitalRead(A4);
if (state == LOW){
blue();
digitalWrite(13,HIGH);
}
else
{
 esp8266();
}
}


void blue() {
 if (bluetooth.available()) {
   char receivedChar = bluetooth.read();
   Serial.print("Received: ");
   Serial.println(receivedChar);

   switch (receivedChar) {
     case '1':
       analogWrite(fan_pin,250); // Turn relay 1 ON
       Serial.println("FAN ON");

       break;
     case '2':
       digitalWrite(fan_pin, LOW);  // Turn relay 1 OFF
       Serial.println("FAN OFF");
```

```
    break;
  case '3':
    digitalWrite(light_pin, HIGH); // Turn relay 2 ON
    Serial.println("LIGHT ON");

    break;
  case '4':
    digitalWrite(light_pin, LOW);  // Turn relay 2 OFF
    Serial.println("LIGHT OFF");

    break;
  default:
    Serial.println("Invalid command");

    break;
 }
}
```

6.2 Sample Screen Shots

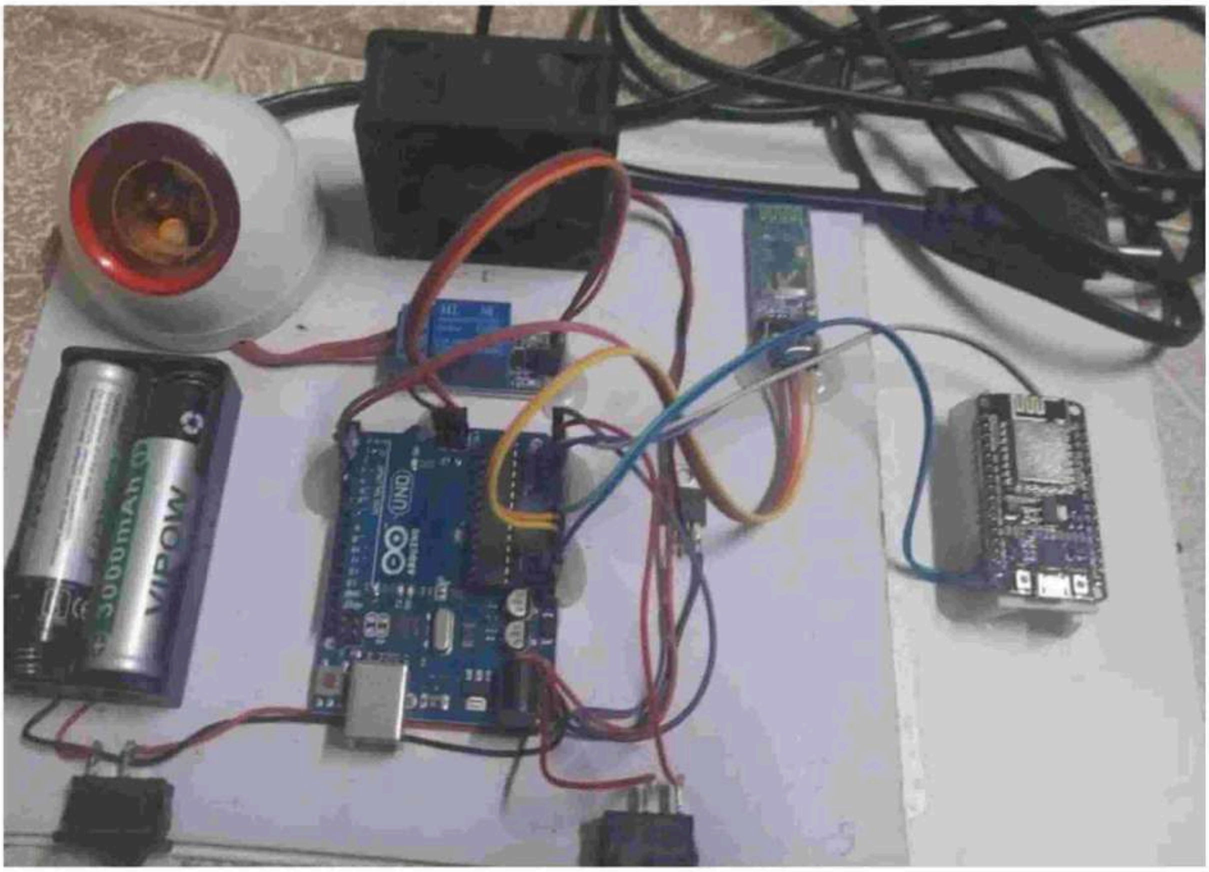

Fig.6.2.1 Hardware Setup

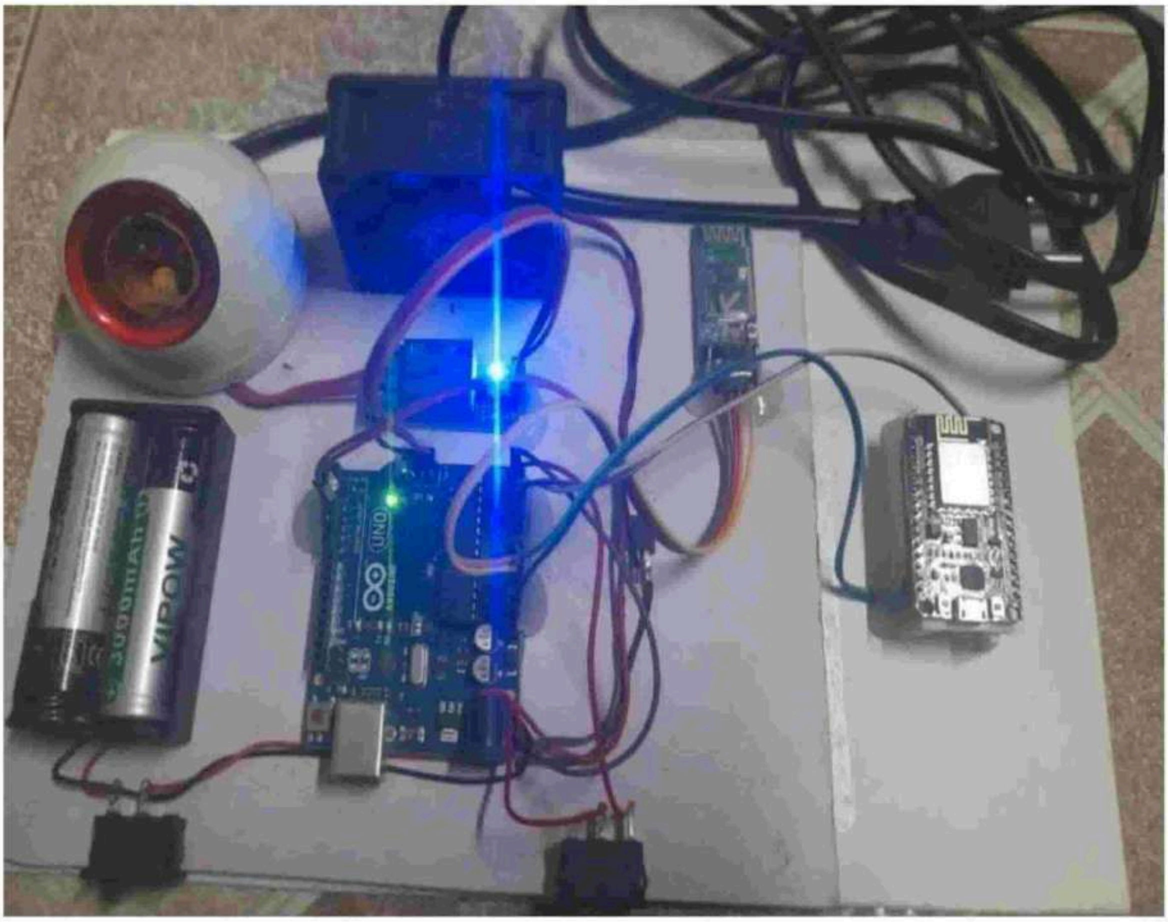

Fig.6.2.2 Working Of Project

Blynk cloud:

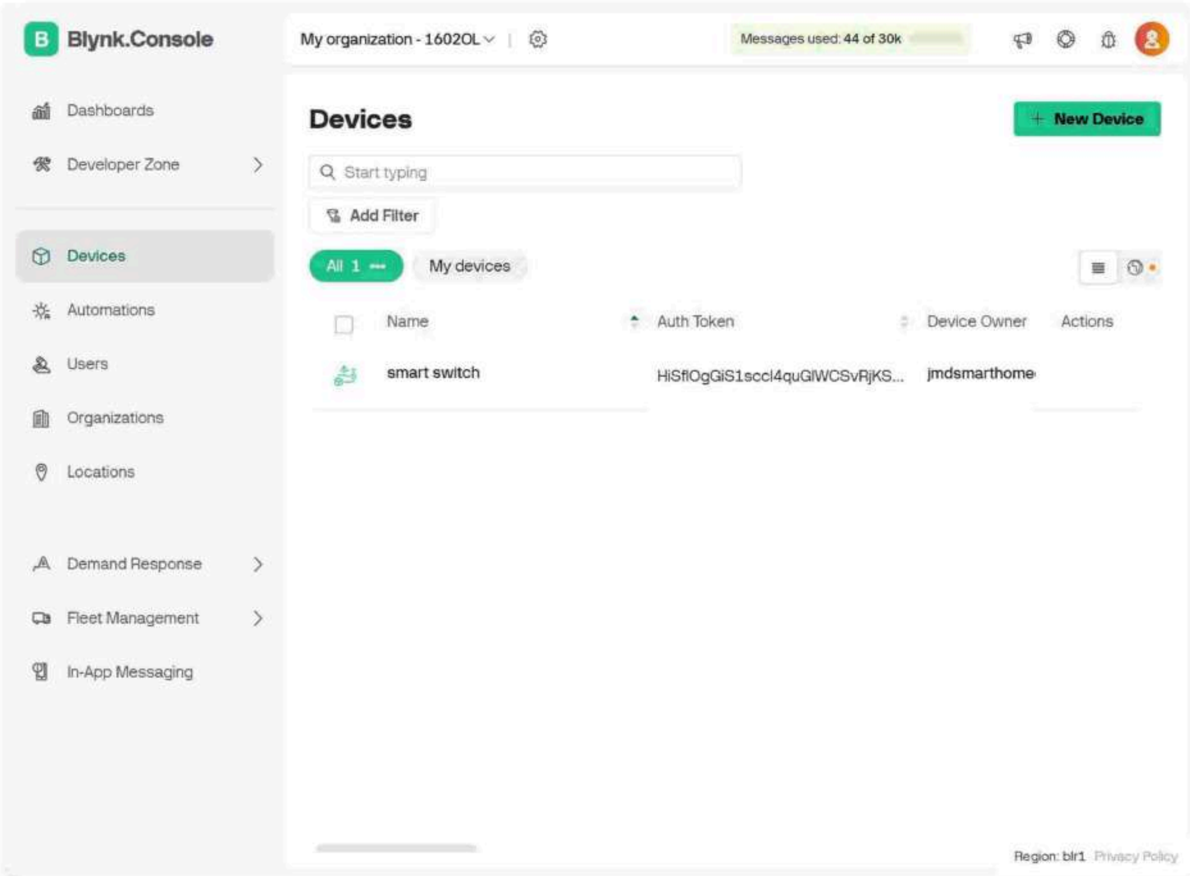

Fig.6.2.3 Accessing Into Blynk Cloud

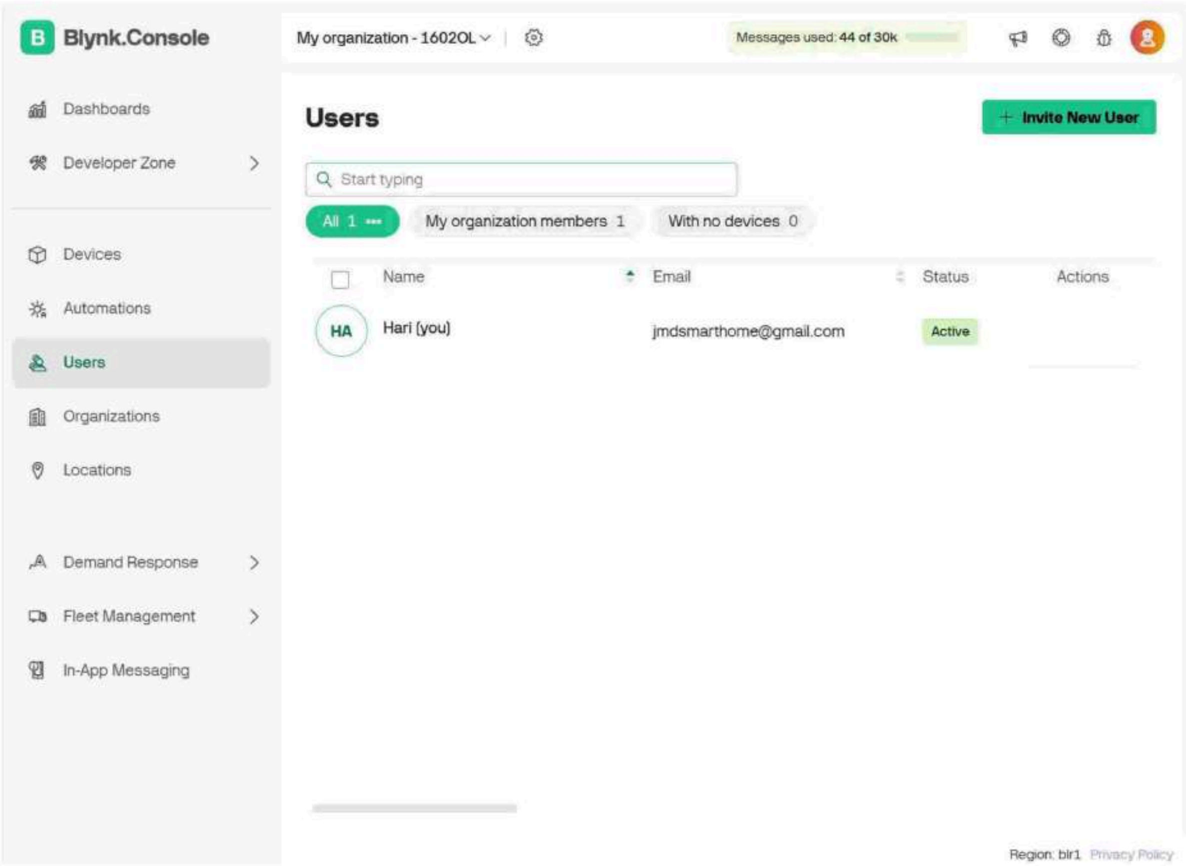

Fig.6.2.4 Scanning The Number Of Users

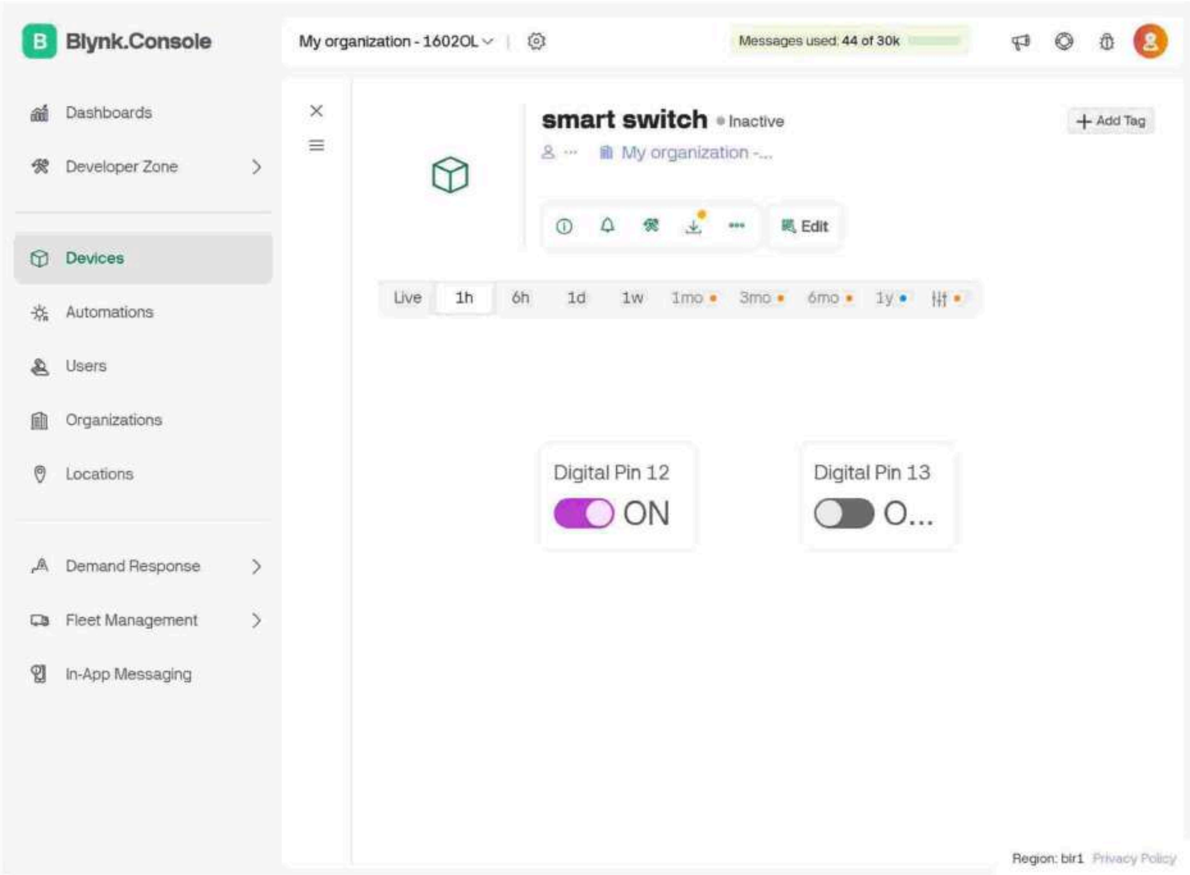

Fig.6.2.5 Giving Commands

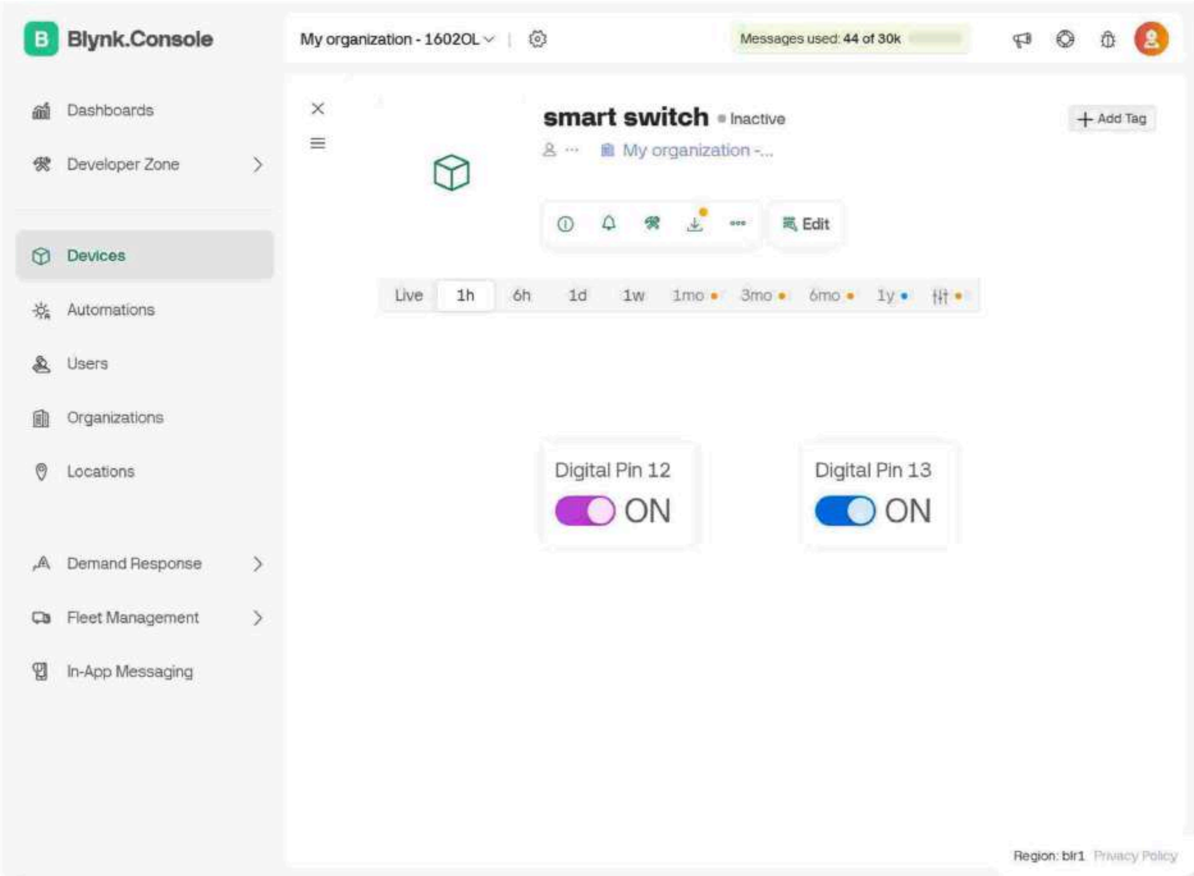

Fig.6.2.7 Device Controlled From Anywhere

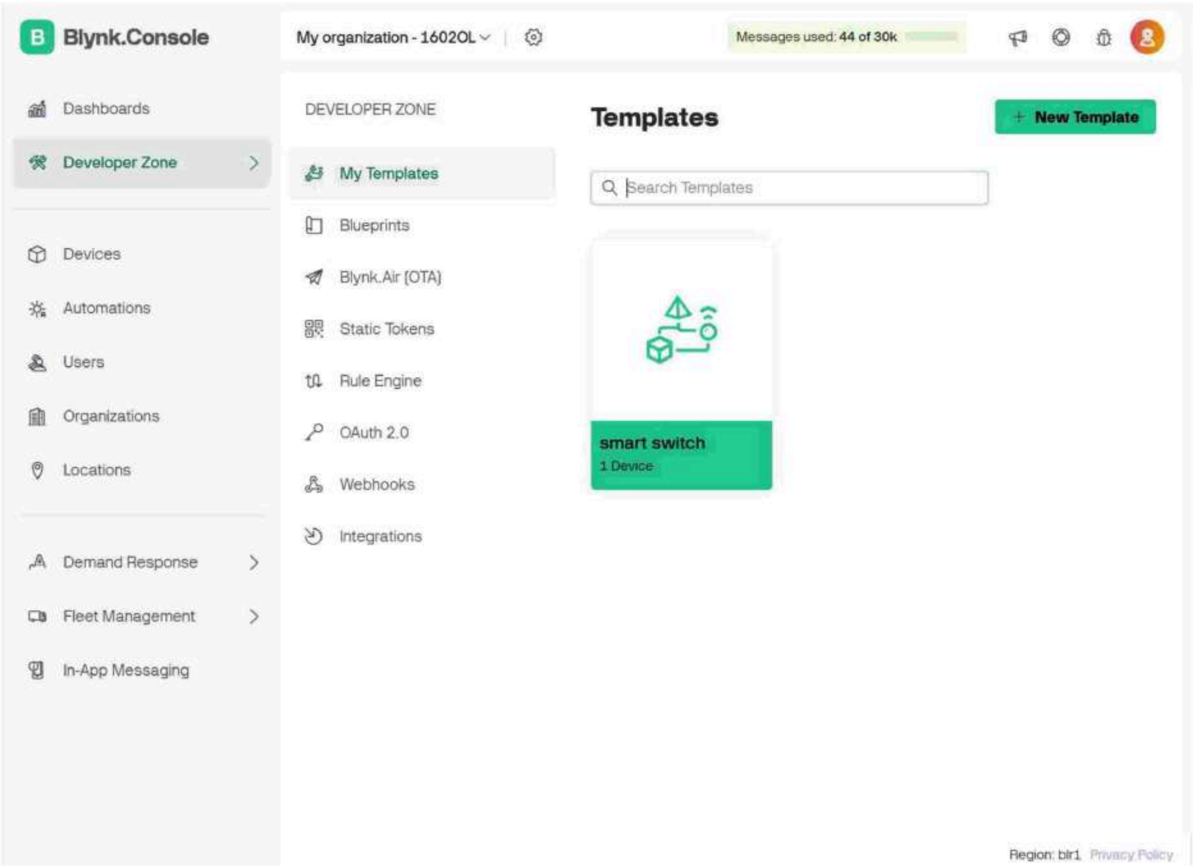

Fig.6.2.8 Accessing Into Templates

CHAPTER 7

COST ANALYSIS

S.No.	**Component**	**Quantity**	**Unit Price (INR)**	**Total (INR)**
1.	Arduino Uno	1	550	550
2.	Esp 8266	1	250	250
3.	HC-05 Bluetooth module	1	230	230
4.	Light	1	30	30
5.	Relay module	1	50	50
6.	switch	2	5	10
7.	Light Holder	1	50	50
8.	Fan	1	100	100
9.	Jumper Wires	3 Sets	20	60
10.	Battery	2	230	460

Total cost=1,790

Table 7.1 Cost analysis

CHAPTER 8

SUMMARY

This project report deals with the design and development of a Smart Home Automation System for remote and local control of home electric appliances using Wi-Fi (ESP8266), Bluetooth (HC-05), and the Blynk Cloud platform. The system offers more convenience to the user, security, and energy saving by controlling appliances such as lights and fans remotely using a mobile app and manual switches too. The initial objective of the project is to develop low-cost, flexible, and scalable automation hardware that will easily be installed in small office or home environments without radically changing buildings or spending large amounts of capital on equipment.

The system architecture provides a number of control options. Wi-Fi control is gained via the usage of the ESP8266 Wi-Fi module that communicates with the Blynk Cloud, sending instructions to manage devices remotely using a smartphone application by an individual. Bluetooth local control is achieved via the usage of the HC-05 module to provide control locally in situations of no connection to the internet. EEPROM is used for data storage of a wake word. There are also switch physical ones for the sake of simulating a network failure or for those used to the old switches. The Arduino Uno board is the core of the system, which processes the input from the mobile app or switches and turns on/off the corresponding relay modules.

Hardware materials of the project include ESP8266, HC-05, relay module, switches, LED or light, fans, battery or DC power adapter, and jumper wires for joining. The software development environment employed is the Arduino IDE for microcontroller programming and upload and Blynk smartphone app, with a simple graphical user interface for controlling and displaying live. The system is shown to function well and with capability to remotely and locally manage devices and pre configure it for the physically disabled or elderly customer and home improvement.

The life cycle development was constructed step by step from planning to hardware build-out, software build-out, integration, and test. Milestones were set and achieved on a 10-week schedule that shows timely delivery and development. Minimum compromise and sacrifices were taken care of while considering design aspects such as cloud potential internet dependence, Bluetooth range limitation, and power optimization. ESP32 (where Wi-Fi and Bluetooth are integrated) was also considered but not utilized with a motive of cost reduction.

In short, the smart home automation project is an affordable and feasible alternative to other home control systems. It's affordable, easy to install, and highly flexible, thus making it a feasible solution for actual applications, especially in regions where up to now smart infrastructure is yet to be developed.

CHAPTER 9

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